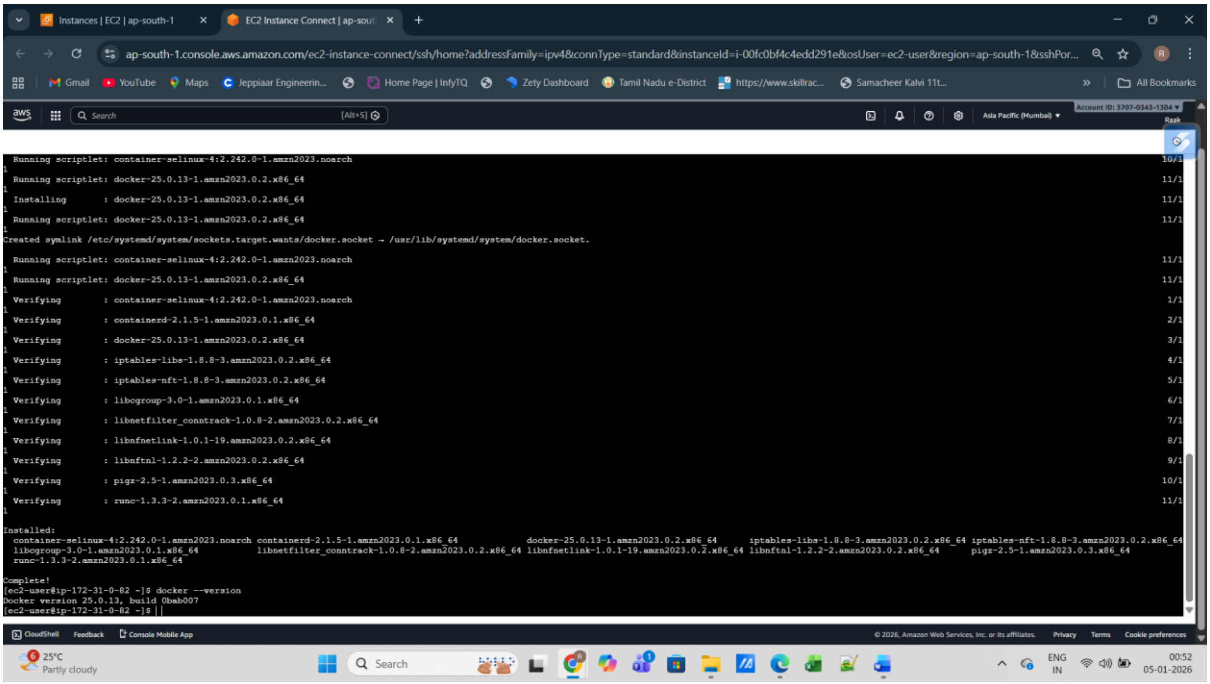
WEEK 8 ASSIGNMENT: Dockerize a Web Application

1. Installed docker and version check:



2. Simple **python** code:

```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Dockerized Python App running on EC2 "

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Requirements:
Flask

3. Muti-stage build docker file:

    # -------- Stage 1: Build --------

    FROM python:3.10-slim AS builder

    WORKDIR /app

    COPY requirements.txt .

    RUN pip install --user -r requirements.txt

    # -------- Stage 2: Runtime --------

    FROM python:3.10-slim

    WORKDIR /app

    COPY --from=builder /root/.local /root/.local

    COPY app.py .

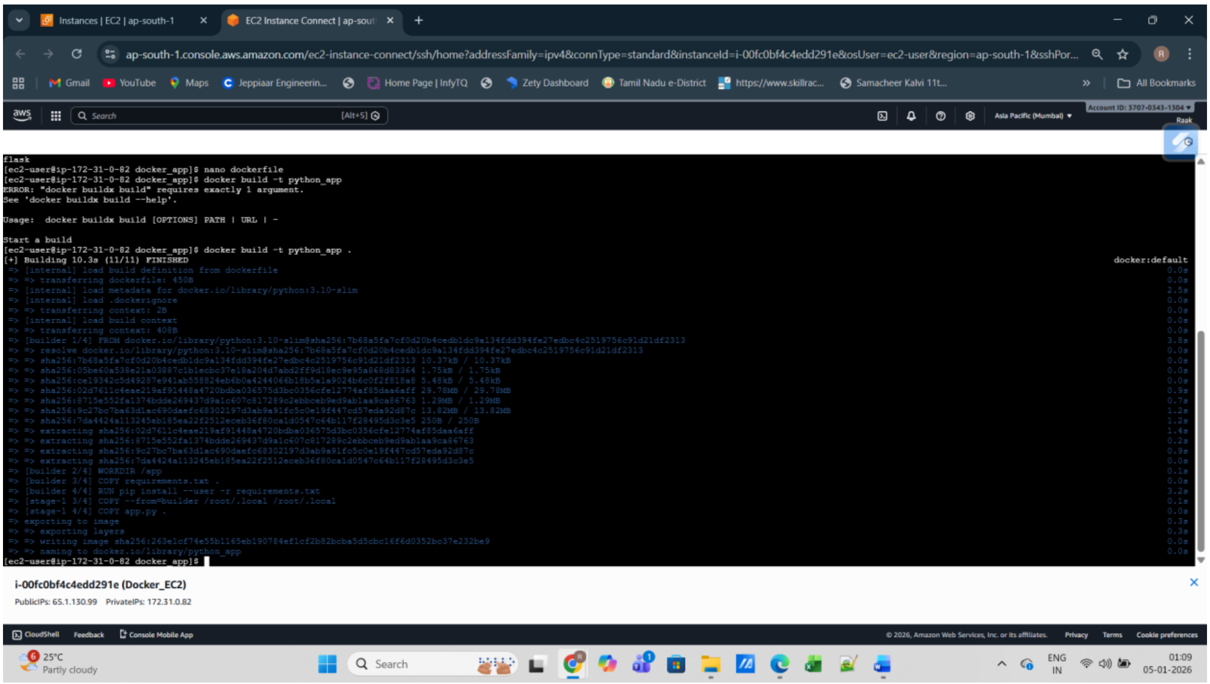
    ENV PATH=/root/.local/bin:$PATH

    EXPOSE 5000

    CMD ["python", "app.py"]

Image build with below command:

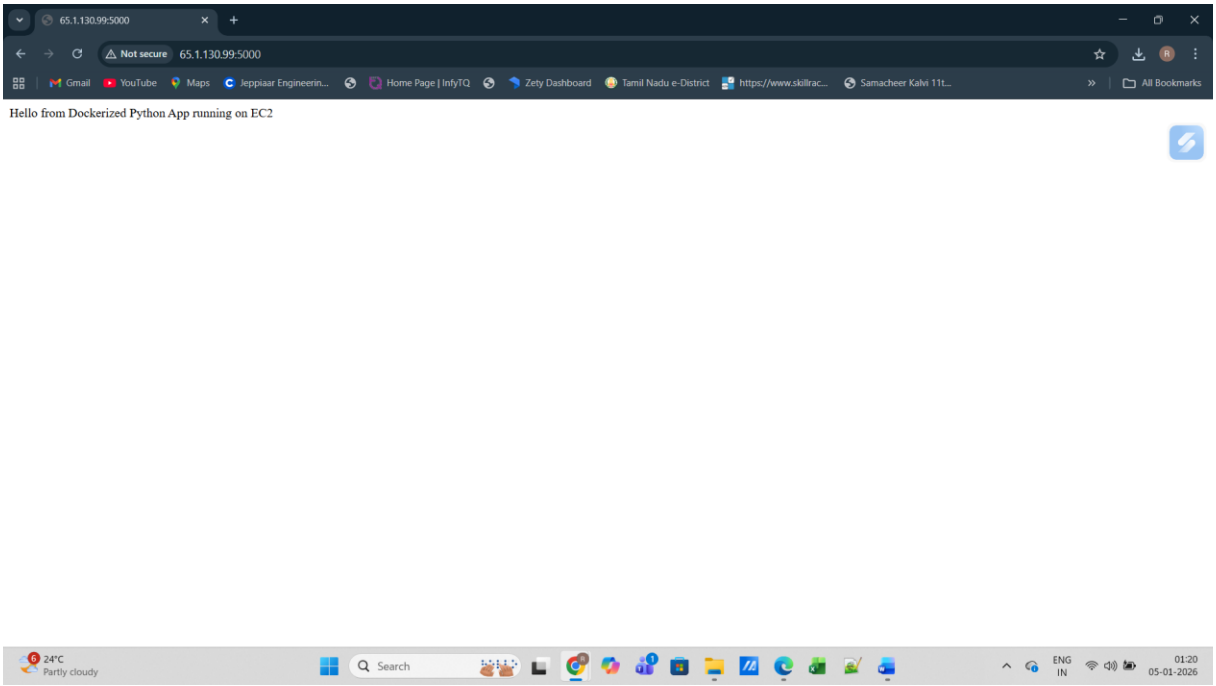
Docker build -t python_app .



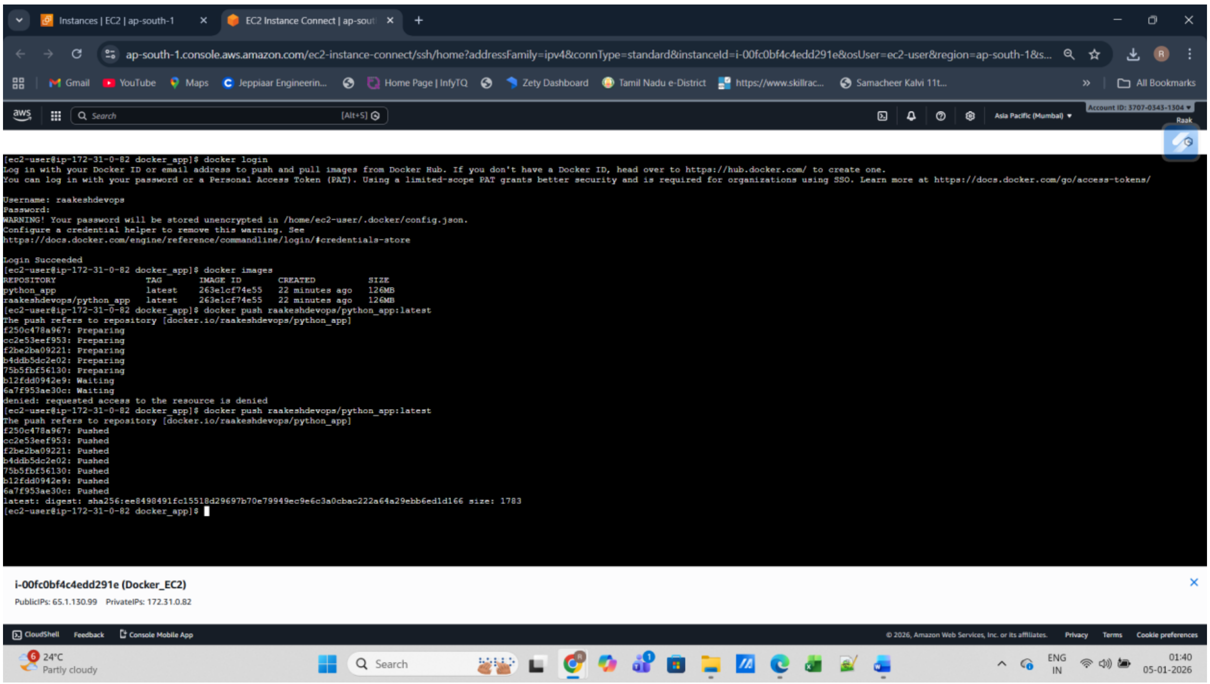
4. Running container and browser output:

```
[ec2-user@ip-172-31-0-82 docker_app]$ docker images
REPOSITORY   TAG      IMAGE ID       CREATED         SIZE
python_app   latest   263e1cf74e55   2 minutes ago   126MB
[ec2-user@ip-172-31-0-82 docker_app]$ docker run -d -p 5000:5000 --name container1 python_app
c446b8bf8f389b96c2d45990733ee999a91baa9a4586e1e2765ff9f3fe7540da
[ec2-user@ip-172-31-0-82 docker_app]$ docker ps
CONTAINER ID   IMAGE        COMMAND            CREATED         STATUS         PORTS                                       NAMES
c446b8bf8f38   python_app   "python app.py"    4 minutes ago   Up 4 minutes   0.0.0.0:5000->5000/tcp, :::5000->5000/tcp   container1
[ec2-user@ip-172-31-0-82 docker_app]$
```

Browser Output:



5. Pushing images to docker HUB.



Docker hub image link:

https://hub.docker.com/repositories/raakeshdevops